

sk

pk

query q

① Send encrypted query $\llbracket q \rrbracket$

pk

$\llbracket x \rrbracket = (\llbracket x_1 \rrbracket, \dots, \llbracket x_n \rrbracket)$

② Homomorphically evaluate
 $\llbracket q \rrbracket$ on $\llbracket x \rrbracket$

③ Return index $\llbracket i^* \rrbracket$ and data $\llbracket x_{i^*} \rrbracket$

④ Decrypt i^* and x_{i^*}

Client

Server